

How a clownfish swims

the biomechanics behind the aquarium level's fish rig · *Amphiprion ocellaris* · numbers, formulas and how far to trust each one

The level is $\times 10$ in **space** and real in **time**: speeds are in body lengths per second, frequencies in hertz. The fish is 3.5 in (8.9 cm) real, 0.889 m in the level; cruise 3.048 m/s (3.43 BL/s).

A · Body wave: amplitude grows from head to tail; wavelength $\approx$ one body length



eight phases of one beat: 0, $\frac{1}{8}$... $\frac{7}{8}$

solid fill: phase 0 · dashed outline: phase $\frac{1}{4}$

$$y(s, t) = A(s) \cdot L \cdot \sin[2\pi (s/\lambda - f \cdot t)]$$

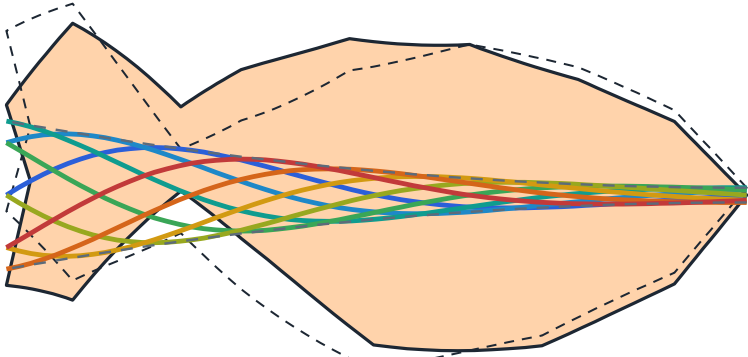
$$A(s) = (A/2) \cdot (0.1 + 0.9 s^2), \lambda = L$$

$s = 0$ at the nose, 1 at the tail tip

ours · envelope shape: drawing only

unverified · $A \approx 0.2 L$, $\lambda \approx L$: widely cited

$A \approx 0.2 L$
peak to peak



swimming direction

wave travels head $\rightarrow$ tail

$L =$ body length: 3.5 in (8.9 cm) real · 0.889 m at $\times 10$

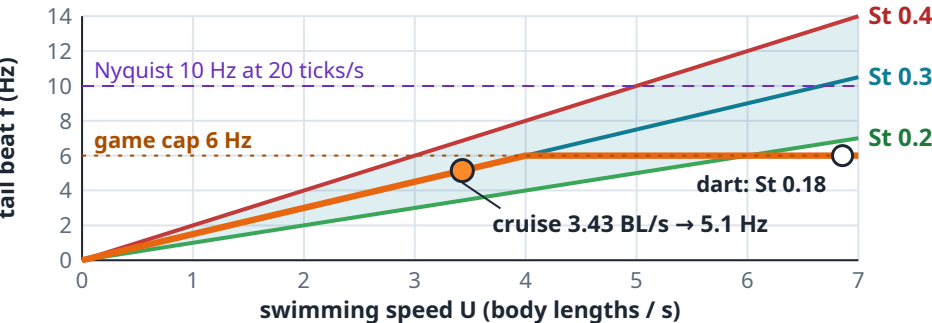
B · Tail-beat frequency follows speed (Strouhal number)

$$f = St \cdot U / A, A = 0.2 L \Rightarrow f = 5 \cdot St \cdot U / L$$

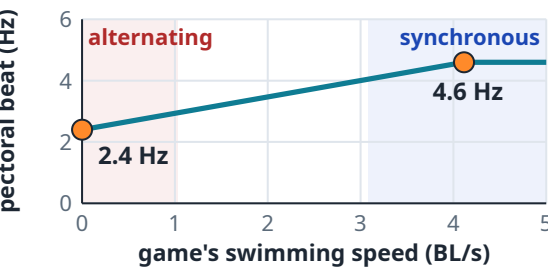
verified · St 0.2-0.4 in fish

unverified · $A = 0.2 L$

ours · $St = 0.3$, cap



D · Pectoral fins: beat rate and gait



verified · *A. ocellaris* 2.4 $\rightarrow$ 4.6 /s, surge trials

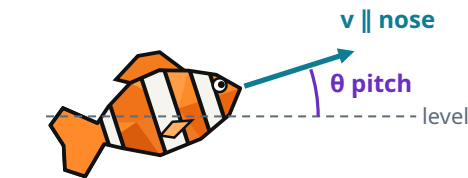
other species · gait switch: hint

slow: alternating, 2.4 Hz

fast: synchronous, 4.6 Hz

0.9 s shown

F · Facing, velocity along the nose, and the control chain



ψ yaw = ROTATION_C
 θ pitch = $-\text{ROTATION}_B$, limit 40°
 ϕ roll = ROTATION_A, bank $\leq 15^\circ$

the physics hull never turns;
only the visible parts do

ours · every box: the game

unverified · layering idea: Tu & Terzopoulos

buttons $\rightarrow$ wanted direction d

target facing ψ^* , θ^* (pitch $\leq 40^\circ$)

yaw, pitch springs (ω , ζ) · bank

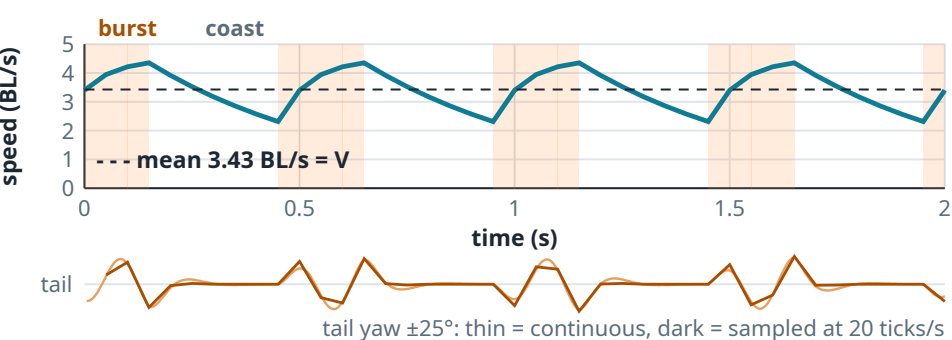
burst-and-coast speed U

$$v = U \cdot (\cos\theta \cos\psi, \cos\theta \sin\psi, \sin\theta)$$

XSPEED · YSPEED · ZSPEED

C · Burst-and-coast: the gait the game gives the fish

cycle 0.5 s: burst 40 % (tail beats) · coast 60 % (body straight)



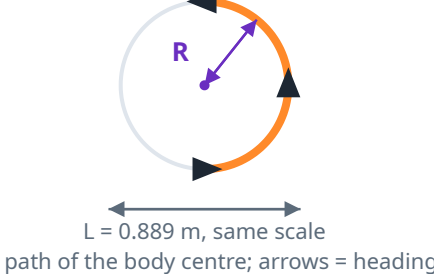
verified · drag $\approx 4 : 1$, $\approx 45\%$ saved (koi)

ours · cycle, share, τ

E · Turning: the head leads, the body bends

cruising U-turn: $R \approx 0.4 \text{ m} = 0.44 L$

$$R = V / (2\pi \cdot \omega), V = 3.048 \text{ m/s}, \omega = 1.25 \text{ rev/s}$$



ours · R : game measurement

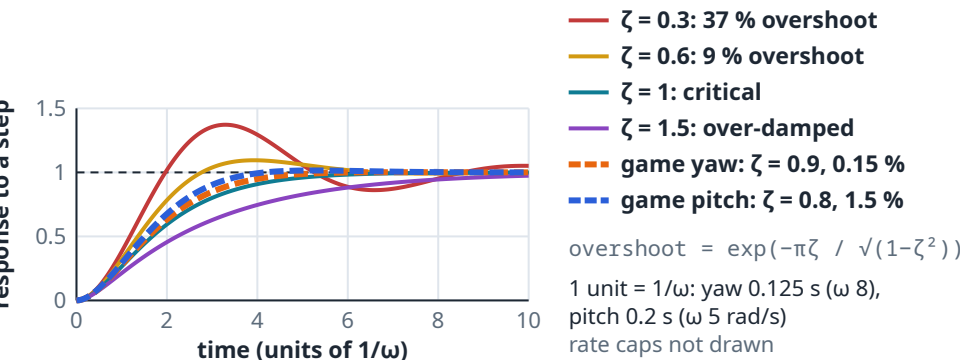
unverified · C-start: Domenici & Blake

escape C-start: four stages



stage 1 bends the body into a C; larger fish bend slower but cover more distance. Definitions, not a clownfish figure.

G · Damped steering: why ζ decides how a turn feels



ours · the game's ζ and ω ; the other four are illustrations

H · Parameter table: every number, its unit, its source and how far to trust it

Quantity	Value	Chip · source	Quantity	Value	Chip · source	Quantity	Value	Chip · source
Body length L	3.5 in (8.9 cm)	ours	Constant cycle idea	qualitative	unverified S6	Pitch ω	5 rad/s	ours
Level scale, space	$\times 10$	ours	Gait cycle	0.5 s	ours	Pitch ζ	0.8	ours
Cruise speed V	3.048 m/s	ours	Burst share	40 %	ours	Pitch rate cap	0.4 rev/s	ours
Strouhal, fish	0.2-0.4	verified S1	Burst speed target	3.99 m/s (1.309 V)	ours	Pitch limit	40°	ours
Strouhal, trout	0.19 $\rightarrow$ 0.22	verified S1	Burst rise τ	0.1 s	ours	Pitch limit, diag.	30°	ours
Strouhal, game	0.3	ours	Coast decay τ	0.5 s	ours	Bank at full yaw	15°	ours
Tail excursion A/L	0.2 peak to peak	unverified S9	Glide τ (released)	0.45 s	ours	Burst dip in turns	up to 25 %	ours
Wavelength λ/L	≈ 1	unverified S9	Pectoral beat, slow	2.4 Hz	verified S4	Wall easing time	0.25 s	ours
Envelope $A(s)$	$0.1 + 0.9 s^2$	ours	Pectoral beat, fast	4.6 Hz	verified S4	U-turn radius R	$\approx 0.4 \text{ m}$ (0.39)	ours
Tail beat at V	5.14 Hz	ours	Speed at fast beat	3.658 m/s (1.2 V)	ours	C-bend gain	0.05 s	ours
Tail-beat cap	6 Hz	ours	Alternating below	0.914 m/s (0.3 V)	other species S5	Body recoil share	20 %	ours
Engine tick rate	20 ticks/s	ours	Synchronous above	2.743 m/s (0.9 V)	other species S5	Tail envelope time	0.06 s	ours
St in a dart	0.18	ours	Damselfish switch	1.87-4.95 BL/s	other species S5	C-start stages	definitions	unverified S7
Dart speed	6.096 m/s	ours	Yaw ω	8 rad/s	ours	Layering idea	idea only	unverified S8
Drag burst : coast	$\approx 4 : 1$ (koi)	verified S3	Yaw ζ	0.9	ours			
Energy saved (koi)	$\approx 45\%$	verified S3	Yaw rate cap	1.25 rev/s	ours			

Sources · live links in the PDF; what could not be verified is marked

S1 Knight 2014, JEB 217:2224

St 0.2-0.4 in fish; trout 0.19-0.22

S2 Nudds et al. 2014, JEB 217:2244

link missing via S1 only

S3 Wu, Yang & Zeng 2007, JEB 210:2181

koi: drag $\approx 4 : 1$; $\approx 45\%$ saved

S4 Marcoux & Korsmeyer 2019, JEB 222(4)

A. ocellaris pectoral 2.4 $\rightarrow$ 4.6 /s, surge

S5 Hale et al. 2006, JEB 209:3708

damselfish, not clownfish: gait switch

S6 Li et al. 2021, Commun. Biol. 4:40

constant cycle idea: not opened

S7 Domenici & Blake 1997, fast-starts

C-start definitions: not opened

S8 Tu & Terzopoulos 1994

layering idea only: not opened

S9 $A \approx 0.2 L$, $\lambda \approx L$: no primary source

link missing widely cited; no source opened

How to read the chips

verified opened; the number is on the page

unverified widely cited, or not opened

ours a game tunable or our own maths

other species another species: a hint only

"ours" values come from the game's code at build time. Not used: Rohr & Fish 2004 (cetaceans). Data dated 2026-09-30.